Yu Hao

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I am a security researcher on the HarmonyOS System Security Team at Huawei, where I lead memory-safety efforts for HarmonyOS. My research spans operating-system security and program analysis for C/C++ systems, combining fuzzing, static analysis, symbolic execution, and LLM-based reasoning to secure real-world kernels such as Linux, Android, and HarmonyOS. Representative work includes automated kernel fuzzing specification generation (SyzDescribe, S&P 23; SyzSpec, CCS 25 Distinguished Paper Award) and LLM-assisted static analysis (E&V; OOPSLA 24). This work has appeared in S&P, USENIX Security, CCS, NDSS, ICSE, FSE, OOPSLA, and TSE, and has produced open-source tools, invited talks, and Linux kernel bug reports and patches.

Education

09/2018 – 09/2024 **Ph.D.** in Computer Science, University of California, Riverside, USA.
Advisor: Zhiyun Qian

09/2015 – 06/2018 **Master** in Control Science and Engineering, Xi'an Jiaotong University, China.
Advisor: Ting Liu

09/2011 – 06/2015 **Bachelor** in Control Science and Engineering, Xi'an Jiaotong University, China.

Experience

10/2024 – present **Security Researcher**
HarmonyOS System Security Team, Huawei, China
HarmonyOS Security, Fuzzing, LLMs for Security, Static Analysis
- Lead the kernel fuzzing technical roadmap for HarmonyOS, targeting memory-safety vulnerabilities in the kernel and system components.
- Lead the Rust technical roadmap for HarmonyOS, eliminating memory-safety bug classes in new system components.
- Apply LLMs to security analysis (AI for Security).

06/2023 – 09/2023 **Research Intern**, Mentor: Weidong Cui
Systems Security and Privacy Research Group, Microsoft Research, USA
LLMs for Security / Program Analysis
- Proposed E&V, an early approach that prompts LLMs to perform static analysis by executing pseudo-code and verifying the results, reducing hallucination without extra tooling.
- Applied E&V to Linux kernel crash triage (locating the function to patch), achieving over 80% accuracy on 170 crashes across seven bug categories.

06/2022 – 09/2022 **Research Intern**, Mentor: Kris Alder
Android Security Team, Google, USA
Linux/Android Kernel Security, Fuzzing, Static Analysis
- Used static analysis to enhance Android kernel fuzzing (internal tooling).

09/2018 – 09/2024 **Graduate Student Researcher**, Advisor: Zhiyun Qian
Center for Research and Education in Cyber Security and Privacy (CRESP), University of California, Riverside, USA
- Built the under-constrained symbolic execution and much of the static-analysis tooling behind the group's kernel security projects listed below.
- Kernel fuzzing: identified the root cause behind the "dependency challenge" (ICSE 22, Google Research Paper Reward); automated syscall specification generation via static analysis (SyzDescribe, S&P 23) and under-constrained symbolic execution (SyzSpec, CCS 25, Distinguished Paper Award); contributed ideas to dependency inference for driver fuzzing (SyzGen++, S&P 24).
- Bug triage & exploitability: co-designed and built the symbolic execution for accurate bisection of fuzzer-exposed vulnerabilities (SymBisect, USENIX Security 24); built the static analysis for exploitability assessment across the Linux ecosystem (SyzBridge, NDSS 24).
- Precise static analysis: built the under-constrained symbolic execution for Linux kernel use-before-initialization detection, eliminating about 87% of false positives (UBITect, FSE 20) and enabling incremental detection (NDSS 22).
- LLM-assisted program analysis: contributed the ideas and static analysis for LLM-integrated static bug detection (FSE 23, OOPSLA 24).

06/2013 – 06/2018 **Research Assistant**, Advisor: Ting Liu
Ministry of Education Key Lab For Intelligent Networks and Network Security, Xi'an Jiaotong University, China
Concurrency Security, Symbolic Execution

Thesis

[PhD] Expanding the Boundaries of OS Kernel Fuzzing Permalink
- [C9, ICSE 22] [C10, S&P 23] [C17, CCS 25, Distinguished Paper Award]

[Master] Research on Malicious Code Hiding Methods Based on Uncertain Interleaving of Concurrent Programs
- [C3, QRS 21, Best Paper Award] [J3, *IEEE Transactions on Reliability*]

Publications

[C]: Conference, [J]: Journal, [P]: Preprint

[C18] NeuroMerge: ML-Guided State Merging for Efficient Symbolic Execution
ISSRE 26 Shenghan Zheng, Shitong Zhu, **Yu Hao**, Xingyu Li, Keyu Man, Zheng Zhang, Qing Deng, Zhiyun Qian, Srikanth Krishnamurthy
IEEE International Symposium on Software Reliability Engineering, ISSRE 26.

[C17] SyzSpec: Specification Generation for Linux Kernel Fuzzing via Under-Constrained Symbolic Execution
CCS 25 **Yu Hao**, Juefei Pu, Xingyu Li, Zhiyun Qian, Ardalan Amiri Sani
ACM SIGSAC Conference on Computer and Communications Security, CCS 25. Tool
- [Distinguished Paper Award]

[C16] SCAD: Towards a Universal and Automated Network Side-Channel Vulnerability Detection
S&P 25 Keyu Man, Zhongjie Wang, **Yu Hao**, Shenghan Zheng, Xin'an Zhou, Yue Cao, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 25. Tool

[C15] SymBisect: Accurate Bisection for Fuzzer-Exposed Vulnerabilities
USENIX Security 24 Zheng Zhang, **Yu Hao**, Weiteng Chen, Xiaochen Zou, Xingyu Li, Haonan Li, Yizhuo Zhai, Zhiyun Qian, Billy Lau
USENIX Security Symposium 2024. Tool
- [Linux Security Summit 25]

[C14] Enhancing Static Analysis for Practical Bug Detection: An LLM-Integrated Approach
OOPSLA 24 Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
ACM SIGPLAN International Conference on Object-Oriented Programming Systems, Languages, and Applications, OOPSLA 24. Tool

[C13] SyzGen++: Dependency Inference for Augmenting Kernel Driver Fuzzing
S&P 24 Weiteng Chen, **Yu Hao**, Zheng Zhang, Xiaochen Zou, Dhilung Kirat, Shachee Mishra, Douglas Schales, Jiyong Jang, Zhiyun Qian
IEEE Symposium on Security and Privacy, S&P 24. Tool

[C12] SyzBridge: Bridging the Gap in Exploitability Assessment of Linux Kernel Bugs in the Linux Ecosystem
NDSS 24 Xiaochen Zou, **Yu Hao**, Zheng Zhang, Juefei Pu, Weiteng Chen, Zhiyun Qian
Network and Distributed System Security Symposium, NDSS 24. Tool

[P1] E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
Yu Hao, Weiteng Chen, Ziqiao Zhou, Weidong Cui
- [AGI Leap Summit 2024] [Symposium on the Science of Security 24]

[C11] Assisting Static Analysis with Large Language Models: A ChatGPT Experiment
FSE 23 Haonan Li, **Yu Hao**, Yizhuo Zhai, Zhiyun Qian
IVR *The ACM International Conference on the Foundations of Software Engineering, Ideas, Visions and Reflections, FSE 23 IVR.* Tool

[C10] SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
S&P 23 **Yu Hao**, Guoren Li, Xiaochen Zou, Weiteng Chen, Shitong Zhu, Zhiyun Qian, Ardalan Amiri Sani
IEEE Symposium on Security and Privacy, S&P 23. Tool Result
- [Linux Security Summit 23] [Qualcomm Product Security Summit 23]
- [Symposium on the Science of Security 24]

[C9] Demystifying the Dependency Challenge in Kernel Fuzzing
ICSE 22 **Yu Hao**, Hang Zhang, Guoren Li, Xingyun Du, Zhiyun Qian, Ardalan Amiri Sani
IEEE/ACM International Conference on Software Engineering, ICSE 22. Tool
- [Google Research Paper Rewards]

[C8] NDSS 22 Progressive Scrutiny: Incremental Detection of UBI bugs in the Linux Kernel
Yizhuo Zhai, **Yu Hao**, Zheng Zhang, Weiteng Chen, Guoren Li, Zhiyun Qian, Chengyu Song, Manu Sridharan, Srikanth V. Krishnamurthy, Trent Jaeger, Paul Yu
Network and Distributed System Security Symposium, NDSS 22. Tool
- [2023 Cyber Security CRA Capstone Poster]

[C7] ACSAC 21 Eluding ML-based Adblockers With Actionable Adversarial Examples
Shitong Zhu, Zhongjie Wang, Xun Chen, Shasha Li, Keyu Man, Umar Iqbal, Zhiyun Qian, Kevin S Chan, Srikanth V Krishnamurthy, Zubair Shafiq, **Yu Hao**, Guoren Li, Zheng Zhang, Xiaochen Zou
Annual Computer Security Applications Conference, ACSAC 21. Tool

[C6] CCS 21 Themis: Ambiguity-Aware Network Intrusion Detection based on Symbolic Model Comparison
Zhongjie Wang, Shitong Zhu, Keyu Man, Pengxiong Zhu, **Yu Hao**, Zhiyun Qian, Srikanth V. Krishnamurthy, Tom La Porta, Michael J. De Lucia
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C5] CCS 21 Statically Discovering High-Order Taint Style Vulnerabilities in OS Kernels
Hang Zhang, Weiteng Chen, **Yu Hao**, Guoren Li, Yizhuo Zhai, Xiaochen Zou, Zhiyun Qian
ACM SIGSAC Conference on Computer and Communications Security, CCS 21. Tool

[C4] FSE 20 UBITect: A Precise and Scalable Method to Detect Use-before-Initialization Bugs in Linux Kernel
Yizhuo Zhai, **Yu Hao**, Hang Zhang, Daimeng Wang, Chengyu Song, Zhiyun Qian, Mohsen Lesani, Srikanth V. Krishnamurthy, Paul Yu
ACM SIGSOFT International Symposium on Foundations of Software Engineering, FSE 20. Tool
- [2023 Cyber Security CRA Capstone Poster]

[J3] ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE Transactions on Reliability. Code Result

[C3] QRS 21 ConcSpectre: Be Aware of Forthcoming Malware Hidden in Concurrent Programs
Yang Liu, Ming Fan, Ting Liu, **Yu Hao**, Zisen Xu, Kai Chen, Hao Chen, and Yan Cai
IEEE International Conference on Software Quality, Reliability, and Security, QRS 21. Code Result
- [Best Paper Award]

[J2] TSE Tell You a Definite Answer: Whether Your Data is Tainted During Thread Scheduling
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Ting Liu
*IEEE Transactions on Software Engineering, **TSE**.* Tool Benchmarks Result
- [S&P 17 Poster] Patent: [PCT] [CN]

[J1] Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ting Liu
IEEE Access. Tool

[C2] ICST 17 Automated Testing of Definition-Use Data Flow for Multithreaded Programs
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, Pei Liu, Jialiang Chang, **Yu Hao**, Ting Liu
IEEE International Conference on Software Testing, Verification and Validation, ICST 17. Tool

[C1] SATE 17 Debugging Multithreaded Programs as if They Were Sequential
Xiaodong Zhang, Zijiang Yang, Qinghua Zheng, **Yu Hao**, Pei Liu, Lechen Yu, Ming Fan, Ting Liu
IEEE International Conference on Software Analysis, Testing and Evolution, SATE 16. Tool

Patent

WO2017181628 - TAINT ANALYSIS METHOD EMPLOYING SYMBOLIC COMPUTATION AND USED FOR DYNAMIC PARALLEL PROGRAM [PCT] [CN]

Presentations, Talks

- E&V: Prompting Large Language Models to Perform Static Analysis by Pseudo-code Execution and Verification
  - *AGI Leap Summit 2024* Website
  - *Symposium on the Science of Security, HoTSoS 2024* Website
- SyzDescribe: Principled, Automated, Static Generation of Syscall Descriptions for Kernel Drivers
  - *Linux Security Summit North America 2023* Website Slides Recording
  - *Qualcomm Product Security Summit 2023* Website
  - *44th IEEE Symposium on Security and Privacy, S&P 2023* Website Recording
  - *Symposium on the Science of Security, HoTSoS 2024* Website

● Demystifying the Dependency Challenge in Kernel Fuzzing
- *44th IEEE/ACM International Conference on Software Engineering, ICSE 2022* Website Slides Recording

Honors, Awards

2025	CCS 2025, Distinguished Paper Award
2024	Laxmi Bhuyan Fellowship Award, University of California, Riverside
	Dissertation Completion Fellowship Award, University of California, Riverside
2023	IEEE S&P Student Travel Grant
	Google Research Paper Rewards
	The Linux Foundation's Travel Fund
2021	QRS 2021 Best Paper Award
2018	Dean's Distinguished Fellowship, University of California, Riverside
	Outstanding Master's Graduate, Xi'an Jiaotong University
2013	Second Prize, Xi'an JiaoTong University ACM-ICPC Competition
	Second Prize, Xi'an JiaoTong University MCM Competition
2010	First Prize, Mathematical Olympiad in Provinces, Inner Mongolia
	Third Prize, Physics Olympiad in Provinces, Inner Mongolia
	Third Prize, Chemistry Olympiad in Provinces, Inner Mongolia
2007	Gold Medal, National Mathematics Olympiad

Academic Service

Program Committee	CCS 2027, 2026 RAID 2026 FORGE@ICSE 2025, 2024 MSR 2024 (Junior PC)
Reviewer	TIFS 2026, 2024, 2023 TDSC 2026 TOSEM 2026, 2025 Computing Surveys 2026 EAI SecureComm 2023
Sub-Reviewer	S&P 2025, 2024, 2021, 2020 CCS 2024 USENIX Security 2025, 2021 NDSS 2025, 2021, 2020
AEC	EuroSys 2023 ISSTA 2024 ECOOP 2024

References

Advisor of Ph.D. Degree	**Zhiyun Qian** *Everett and Imogene Ross Professor* Computer Science and Engineering department, University of California, Riverside, USA Contact: zhiyunq@cs.ucr.edu
Collaborator	**Ardalan Amiri Sani** *Associate Professor* Computer Science Department, University of California, Irvine, USA Contact: ardalan@uci.edu
Collaborator	**Srikanth V. Krishnamurthy** *Professor, IEEE Fellow* Computer Science and Engineering department, University of California, Riverside, USA Contact: krish@cs.ucr.edu
Intern Mentor at Microsoft Research	**Weidong Cui** *Partner Research Manager* Systems Security and Privacy Research Group, Microsoft Research, Redmond, USA Contact: Weidong.Cui@microsoft.com

Advisor of Master Degree

Ting Liu *Professor, Associate Dean*
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